Homework #0

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A1.

$$TP = 0.99 \cdot \frac{1}{10{,}000}$$

$$FP = 0.01 \cdot \frac{9{,}999}{10{,}000}$$

$$\begin{aligned}\frac{TP}{TP+FP} &= \frac{0.99 \cdot \frac{1}{10{,}000}}{0.99 \cdot \frac{1}{10{,}000} + 0.01 \cdot \frac{9{,}999}{10{,}000}} \\ &= \frac{1}{102}\end{aligned}$$

The chances that you actually have disease is $\frac{1}{102}$.

A2.

(a)

$$\begin{aligned}&1.\ \mathrm{Var}[X_i] = \mathbb{E}\left[X_i^2\right] - (\mathbb{E}[X_i])^2 \\ &2.\ \mathrm{Cov}(X_i, X_i) = \mathrm{Var}[X_i] = \Sigma_{i,i} \\ &\Rightarrow \mathbb{E}\left[X_i^2\right] = \Sigma_{i,i} + \mu_i^2\end{aligned}$$

(b)

$$\begin{aligned}\mathbb{E}\left[X^\top X\right] &= \mathbb{E}\left[X_1^2 + X_2^2 + \cdots + X_n^2\right] \\ &= \mathbb{E}\left[\sum_{i=1}^n X_i^2\right] \\ &= \sum_{i=1}^n \mathbb{E}\left[X_i^2\right] \\ &= \sum_{i=1}^n (\Sigma_{i,i} + \mu_i^2) \\ &= \sum_{i=1}^n \Sigma_{i,i} + \sum_{i=1}^n \mu_i^2 \\ &= \mathrm{tr}(\Sigma) + \mu^\top \mu\end{aligned}$$

A3.

(a)

$$\begin{aligned}\mathbb{E}[aX_1 + b] &= a\,\mathbb{E}[X_1] + b \\ &= a\mu + b \\ &= 0\end{aligned}$$

$$
\begin{aligned}
\operatorname{Var}[aX_1+b] &= \mathbb{E}\big[(aX_1+b)^2\big] - (\mathbb{E}[aX_1+b])^2 \\
&= \mathbb{E}\big[a^2X_1^2 + abX_1 + b^2\big] \\
&= a^2\,\mathbb{E}\big[X_1^2\big] + ab\,\mathbb{E}[X_1] + b^2 \\
&= a^2(\operatorname{Var}[X_1] + (\mathbb{E}[X_1])^2) + ab\mu + b^2 \\
&= a^2(\sigma^2+\mu^2) + ab\mu + b^2 \\
&= a^2\sigma^2 + a^2\mu^2 + ab\mu + b^2 \\
&= a^2\sigma^2 + (a\mu+b)^2 \\
&= a^2\sigma^2 \\
&= 1
\end{aligned}
$$

$$
\Rightarrow a = \pm\frac{1}{\sigma}, \quad b = \mp\frac{\mu}{\sigma}
$$

(b) Let $Z = aX_1 + b \sim \mathcal{N}(0,1)$. The pdf of Z: $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$.

$$
\begin{aligned}
\mathbb{E}[|aX_1+b|] &= \mathbb{E}[|Z|] \\
&= \int_{-\infty}^{\infty} |z| \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\mathrm{d}z \\
&= 2\int_0^{\infty} z\frac{1}{\sqrt{2\pi}} e^{-z^2/2}\mathrm{d}z \\
&= \sqrt{\frac{2}{\pi}}\int_0^{\infty} z e^{-z^2/2}\mathrm{d}z
\end{aligned}
$$

Let $u = \frac{z^2}{2} \Rightarrow \mathrm{d}u = z\mathrm{d}z$:

$$
\begin{aligned}
\sqrt{\frac{2}{\pi}}\int_0^{\infty} e^{-u}\mathrm{d}u &= \sqrt{\frac{2}{\pi}}\big[-e^{-u}\big]_0^{\infty} \\
&= \sqrt{\frac{2}{\pi}}(0-(-1)) \\
&= \sqrt{\frac{2}{\pi}}
\end{aligned}
$$

(c)

$$
\begin{aligned}
\mathbb{E}[X_1+2X_2] &= \mathbb{E}[X_1] + 2\,\mathbb{E}[X_2] \\
&= \mu + 2\mu \\
&= 3\mu \\
\operatorname{Var}[X_1+2X_2] &= \mathbb{E}\big[(X_1+2X_2)^2\big] - (\mathbb{E}[X_1+2X_2])^2 \\
&= \mathbb{E}\big[X_1^2 + 4X_1X_2 + 4X_2^2\big] - (3\mu)^2 \\
&= \mathbb{E}\big[X_1^2\big] + 4\,\mathbb{E}[X_1]\,\mathbb{E}[X_2] + 4\,\mathbb{E}\big[X_2^2\big] - 9\mu^2 \\
&= (\sigma^2+\mu^2) + 4\mu^2 + 4(\sigma_2+\mu^2) - 9\mu^2 \\
&= 5\sigma^2
\end{aligned}
$$

(d)

$$\begin{aligned}
\mathbb{E}\big[\sqrt{n}(\hat{\mu}_n - \mu)\big] &= \sqrt{n}(\mathbb{E}[\hat{\mu}_n] - \mu) \\
&= \sqrt{n}\left(\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] - \mu\right) \\
&= \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] - \mu\right) \\
&= \sqrt{n}\left(\frac{1}{n}\cdot n\mu - \mu\right) \\
&= \sqrt{n}(\mu - \mu) = 0
\end{aligned}$$

$$\begin{aligned}
\operatorname{Var}\big[\sqrt{n}(\hat{\mu}_n - \mu)\big] &= n\operatorname{Var}[\hat{\mu}_n] \\
&= n\operatorname{Var}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] \\
&= n\cdot\frac{1}{n^2}\sum_{i=1}^{n}\operatorname{Var}[X_i] \\
&= \frac{1}{n}\cdot n\sigma^2 = \sigma^2
\end{aligned}$$

A4.

(a)

$$\begin{bmatrix} 0 & 1 & -1 & -2 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 2 \end{bmatrix}\begin{bmatrix} -1 & 2 & 0 \\ -2 & 1 & 2 \\ 2 & 0 & -3 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 3 & 1 \\ -2 & 2 & 1 \\ 1 & -1 & -1 \end{bmatrix}$$

(b) $\operatorname{rank}(AB) = 3$ (AB has 3 non-zero, independence rows).

Since $\operatorname{rank}(AB) \le \min(\operatorname{rank}(A), \operatorname{rank}(B))$, and $\operatorname{rank}(A) \le \min(3,4) = 3$, $\operatorname{rank}(B) \le \min(4,3) = 3$, we have $\operatorname{rank}(A) = \operatorname{rank}(B) = 3$.

(c) AB is invertible $\Rightarrow C = (AB)^{-1}$

$$\begin{aligned}
\det(AB) &= (-2)\begin{vmatrix} 2 & 1 \\ -1 & -1 \end{vmatrix} - 3\begin{vmatrix} -2 & 1 \\ 1 & -1 \end{vmatrix} + 1\begin{vmatrix} -2 & 2 \\ 1 & -1 \end{vmatrix} \\
&= (-2)(-2+1) - 3(2-1) + 1(2-2) \\
&= 2 - 3 + 0 = -1
\end{aligned}$$

$$\operatorname{adj}(AB) = \begin{bmatrix} -1 & -1 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}^{\top} = \begin{bmatrix} -1 & 2 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

$$C = \frac{1}{\det(AB)}\operatorname{adj}(AB) = -1\cdot\begin{bmatrix} -1 & 2 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -1 \\ 1 & -1 & 0 \\ 0 & -1 & -2 \end{bmatrix}$$

(d)

$$x = (AB)^{-1}b = Cb = \begin{bmatrix} 1 & -2 & -1 \\ 1 & -1 & 0 \\ 0 & -1 & -2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ -3 \end{bmatrix}$$

A5.

(a)

$$f(x,y) = \sum_{i=1}^{n}\sum_{j=1}^{n} x_i A_{i,j} x_j + \sum_{i=1}^{n}\sum_{j=1}^{n} y_i B_{i,j} x_j + c$$

(b)

$$\frac{\partial f(x,y)}{\partial x_k} = \sum_{j=1}^{n} A_{k,j} x_j + \sum_{i=1}^{n} A_{i,k} x_i + \sum_{i=1}^{n} B_{i,k} y_i$$
$$\frac{\partial f(x,y)}{\partial y_k} = \sum_{j=1}^{n} B_{k,j} x_j$$

(c)

$$\nabla_x f(x,y) = (A + A^\top)x + B^\top y$$
$$\nabla_y f(x,y) = Bx$$

A6.

(a)

$$\begin{bmatrix} v_1 & 0 & 0 \\ 0 & v_2 & 0 \\ 0 & 0 & v_3 \end{bmatrix}^{-1} = \begin{bmatrix} w_1 & 0 & 0 \\ 0 & w_2 & 0 \\ 0 & 0 & w_3 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 1/v_1 \\ 1/v_2 \\ 1/v_3 \end{bmatrix}$$
$$\Rightarrow g(v_i) = \frac{1}{v_i}$$

(b)

$$\begin{aligned} \|Ax\|_2^2 &= (Ax)^\top (Ax) \\ &= x^\top A^\top A x \\ &= x^\top I x \\ &= \|x\|_2^2 \end{aligned}$$

(c) Since B is symmetric, $B = B^\top$. We want to show $(B^{-1})^\top = B^{-1}$.

$$\begin{aligned} BB^{-1} &= I \\ (BB^{-1})^\top &= I^\top \\ (B^{-1})^\top B^\top &= I \\ (B^{-1})^\top B &= I \\ (B^{-1})^\top &= B^{-1} \end{aligned}$$

(d) Let λ be an eigenvalue of C, $\exists v \neq 0$ such that $Cv = \lambda v$.

$$\begin{aligned} v^\top C v &= v^\top (\lambda v) \\ &= \lambda v^\top v \\ &= \lambda \|v\|_2^2 \end{aligned}$$

Since C is positive semi-definite, $v^\top C v \geq 0$. Since $v \neq 0$, $\|v\|_2^2 > 0$.

$$\therefore \lambda = \frac{v^\top C v}{\|v\|_2^2} \geq 0$$

A7.

Part c:

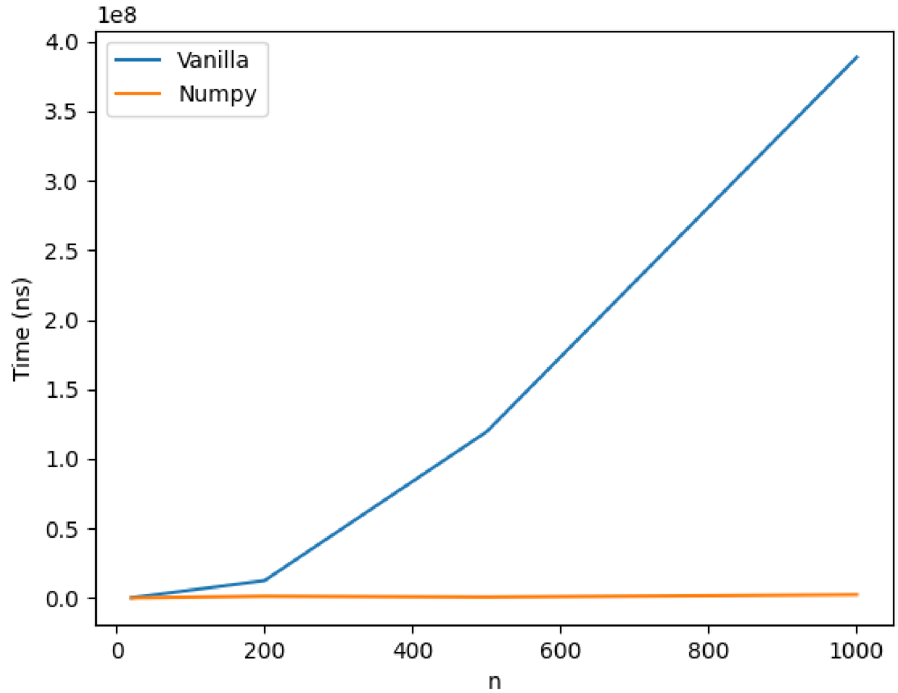

Figure 1: Vanilla vs NumPy

I think the reason for this difference is that the vanilla version uses for-loops, while NumPy is optimized for vectorized operations.

A8.

Part a and b: $n = 40,000$.

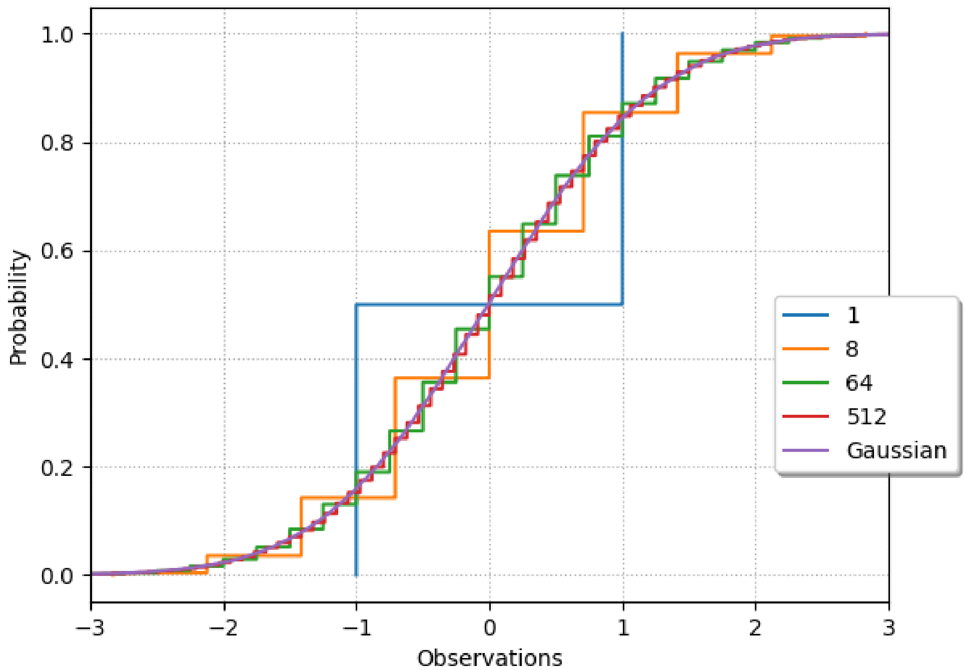

Figure 2: Empirical CDF for $k \in \{1, 8, 64, 512\}$.

As k increases, the empirical CDF of $Y^{(k)}$ converges toward the CDF of the standard normal distribution, which is consistent with the Central Limit Theorem.